

CardFlow

Transaction data as a roadmap for card adoption
in e-commerce & P2P payments

VISA DATASPRINT HACKATHON 2026

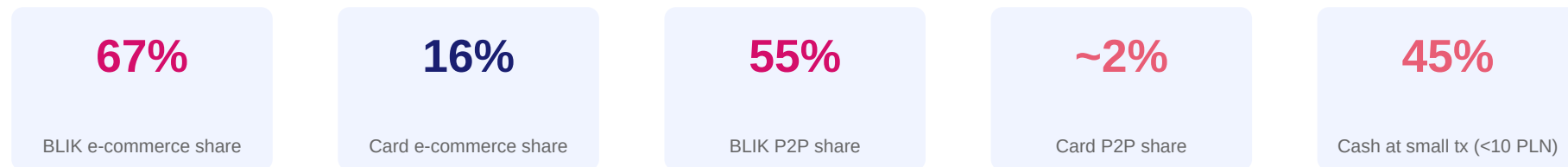
Dane transakcyjne jako mapa drogowa popularyzacji kart w e-commerce i płatnościach P2P

Data sources: Visa (305.5M tx) | GUS | NBP | Gemius

The Problem / Problem

Cards dominate physical POS (58%) but lose in e-commerce (16%) and P2P (~2%).

Karty dominują w POS fizycznym (58%), ale przegrywają w e-commerce (16%) i P2P (~2%).



- The same person taps Visa at a grocery store but uses BLIK online
- BLIK grew from 55% to 67% in e-commerce in just 2 years (2022-2024)
- Cards are losing -2pp/year in online payments
- 25% of household spending (rent, telecom, education) is invisible to cards
- Services (doctors, plumbers, tutors) = 90%+ cash

Data & Methodology / Dane i Metodologia

Source / Zrodlo	Description	Records
Visa Synthetic TX	Anonymized card transactions, Jan 2025 - Jun 2026	305.5M
GUS Household Budget	COICOP spending structure (12 categories)	2024
NBP Payment Stats	Card/cash/BLIK shares, terminal counts	2024
Gemius E-Commerce	Online payment method breakdown	2024
GUS BDL API	Population by region (voivodeship/powiat)	37.5M pop

Methodology: Cross-reference GUS spending structure (COICOP) with Visa MCC categories.

Gap Index = (Visa value share / GUS spending share) x 100. Below 100 = cards underused.

Compliance: All analyses on groups of 30+ cards. 3/75 rule observed. No individual identification.

Key Findings: Card-Free Zones / Strefy bez Kart

Category	GUS %	Visa %	Gap Index	Status
Housing & Utilities	20.6%	0.3%	2	CARD-FREE ZONE
Communications	4.0%	~0%	~1	CARD-FREE ZONE
Education	1.1%	0.03%	3	CARD-FREE ZONE
Healthcare (services)	5.5%	2.5%	46	UNDERUSED
Alcohol & Tobacco	2.5%	0.8%	32	VERY UNDERUSED
Home Furnishings	4.4%	2.6%	59	UNDERUSED
Food & Groceries	27.1%	25.3%	93	WELL COVERED
Restaurants & Hotels	5.7%	9.6%	168	OVERREPRESENTED

~25% of household budgets (housing + telecom) flows through bank transfers — invisible to cards.

E-Commerce: BLIK vs Visa / BLIK vs Visa w E-Commerce

BLIK dominates Polish e-commerce and is growing. Cards are declining.

Method	2022	2023	2024	Trend
BLIK	55%	62%	67%	+12pp in 2 years
Card (Visa/MC)	20%	18%	16%	-4pp in 2 years
Bank Transfer	15%	12%	10%	Declining
Cash on Delivery	8%	6%	5%	Declining

Visa's moat: international subscriptions (Apple, Netflix, Spotify, ChatGPT) = ~8M tx on card rails

E-grocery: 26% of card tx but only 0.12% online. Online avg = 402 vs 131 in-store (3.1x higher).

Projection: at -2pp/year, card share could fall below 10% by 2027 without intervention.

Our Solution: Visa QR Pay / Nasze Rozwiązanie

Every Visa card gets a unique QR code. Scan it to initiate a payment.

No card number shared. Tokenized. Biometric approval. Powered by Visa Direct.

MODE 1: P2P Payment Request

- Friend scans your card's QR code
- Enters amount + description
- You get push notification on phone
- Approve with biometrics (Face ID)
- Money moves via Visa Direct — instant

MODE 2: E-Commerce Checkout

- At checkout, choose 'Visa QR Pay'
- Scan YOUR card with phone/webcam
- Approve push notification (biometric)
- Done — no card number typed ever
- Faster than BLIK, works globally

Use cases: split bills, OLX/Vinted payments, pay the plumber, group collections, online shopping

Visa QR Pay vs BLIK — Comparison

Feature	Traditional Card	BLIK	Visa QR Pay
Steps to pay online	5-7 (type number)	3-4 (code + confirm)	2-3 (scan + approve)
Card number exposed	YES	No (different system)	NO (tokenized)
Works internationally	Yes	Poland only	Yes (global Visa)
P2P payments	Not supported	55% market share	Full support
Works with dead phone	Yes (type number)	No	Yes (QR on card)
Laptop checkout	Type number	Needs phone app	Webcam scans QR
Buyer protection	Full Visa	Limited	Full Visa
Authentication	3D Secure redirect	Bank app confirm	Biometric push

Predictive Models / Modele Predykcyjne

6 models, 3 scenarios, 36-month projection

Model	Method	Key Output
1. Adoption S-Curve	Bass Diffusion (p,q)	7.7M adopters in 3 years (Base)
2. Transaction Volume	Per-user activity model	18.2M tx/month at Y3
3. BLIK Cannibalization	Source attribution	~60% net new to Visa (Base)
4. Revenue & ROI	Fee model + cost structure	Break-even M16, ROI 365%
5. E-Com Market Share	Share simulation	Visa: 8.8% -> 15%+ (Base)
6. Sensitivity	Tornado analysis	Virality (q) = #1 driver

M16

Break-even month

365%

3-year ROI (Base)

149M PLN

3Y Revenue (Base)

~35B PLN

New annual card volume

Recommendations / Rekomendacje

FOR ONLINE MERCHANTS:

- Implement Visa Click to Pay / QR Pay at checkout
- E-grocery: card-first checkout for delivery platforms
- Highlight Visa buyer protection for high-value purchases

FOR BANKS & VISA:

- Launch Visa Direct P2P via QR in banking apps
- Card-on-file for recurring bills (rent, telecom) with cashback
- Tap-to-Phone rollout for service providers (doctors, tradesmen)
- Zero-fee micro-transactions under 10 PLN

FOR CITIES:

- Cashless City pilot (e.g., Krakow) — markets, parking, transport
- Tap-to-Phone for market vendors, food trucks, local services

ROLLOUT PLAN:

- M0-3: Pilot with 2-3 banks, QR in app
- M3-6: QR stickers to all cardholders
- M6-12: Merchant SDK, top 50 e-commerce
- M12-18: QR on all new Visa cards
- M18-24: International rollout (EU)

CardFlow

We turn transaction data into concrete decisions.

We show not just how people pay,
but what to do to make the card their best choice.

Zamieniamy dane transakcyjne w konkretne decyzje.

Pokazujemy nie tylko jak ludzie placa, ale co zrobic,
zeby karta byla najwygodniejszym wyborem.

github.com/mariusz4908/visa-datasprint

Live demo: Streamlit Cloud (link in README)

Visa DataSprint Hackathon 2026 | Team CardFlow